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Magnetic polarity stratigraphy of the Neogene Siwalik Group at Khutia Khola, far western Nepal

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ABSTRACT

The middle Miocene–Pliocene Siwalik Group was deposited in the Himalayan foreland basin in response to uplift and erosion in the Himalayan fold-thrust belt. Results of thermal demagnetization experiments on samples from the Siwalik Group in central and western Nepal demonstrate that laminated siltstones yield paleomagnetic data useful for tectonic and magnetostratigraphic studies. Sandstones and paleosols of the Siwalik Group, however, generally display highly erratic paleomagnetic behavior during thermal demagnetization. On the basis of these observations, siltstones from a well-exposed, 2423-m-thick section of the Siwalik Group in Khutia Khola, far western Nepal, were sampled for magnetic polarity stratigraphy. The Siwalik Group is composed of informal lower, middle, and upper members. Correlation of the resulting polarity stratigraphy with the geomagnetic polarity timescale indicates that the exposed section spans 13.30 to 7.65 Ma. The lower-middle Siwalik boundary occurs at 11.05 Ma, near the beginning of chron C5n. The rate of sediment accumulation increases upsection, similar to rate changes previously observed in the Pakistan Siwalik Group, and probably in response to increasing proximity of the Himalayan thrust belt. In the Khutia Khola section, a discordant declination indicates that this region has rotated about a vertical axis 16.6° counterclockwise with respect to the Indian subcontinent. Measurements of $\delta^{13}C$ in paleosol carbonate indicate the predominance of C_3 plants until 7.65 Ma, and the clear presence of C_4 plants higher in the undated portion of the section.

Keywords: Himalayan orogeny, Siwalik Group, Nepal, Neogene, magnetostratigraphy, geochronology, carbon-13.

INTRODUCTION

The Siwalik Group in the frontal part of the Himalayan fold-thrust belt consists of a several-kilometer-thick prism of sediment shed from the rising Himalayan fold-thrust belt during Miocene-Pliocene time into the Himalayan foreland basin system (Gansser, 1964; Raiverman et al., 1983; Johnson et al., 1986; Tokuoka et al., 1986; Lillie et al., 1987; Acharyya, 1994; Quade et al., 1995; Burbank et al., 1996; DeCelles et al., 1998a). Because of its great potential for elucidating the tectonic and erosional histories of the Himalaya, the Siwalik Group has been the focus of numerous sedimentological, structural, and chronostratigraphic studies. Magnetic polarity stratigraphy, supported by biostratigraphic evidence and fission track ages from rare volcanic tuffs, has been successfully applied as a geochronologic method in the Siwalik Group of northern Pakistan (e.g., Opdyke et al., 1979, 1982; Johnson et al., 1982, 1983, 1986; Tauxe and Opdyke, 1982; Munthe et al., 1983; N. M. Johnson et al., 1985; Raynolds and Johnson, 1985; Cerveny et al., 1988; Burbank et al., 1988; Khan et al., 1988, Friedman et al., 1992; Sangode et al., 1996). In Nepal and parts of northern India, however, the magnetic polarity stratigraphy of the Siwalik Group has been much more sparsely documented, owing in part to the relative remoteness of Siwalik sections and the lesser quality of outcrop. Also lacking in Nepal are volcanic ash beds that might serve to constrain interpretations of the magnetic polarity stratigraphy. However, biostratigraphic evidence from central Nepal (e.g., Munthe et al., 1983; Conroy et al., 1986; Corvinus, 1994) establishes the broad synchroneity of the Nepal Siwalik Group with the well-dated Siwalik Group (14–0.5 Ma) in Pakistan. Several studies of the Siwalik Group in Nepal have produced magnetic polarity stratigraphies (Appel et al., 1991; Harrison et al., 1993; Gautam and Appel, 1994), but the results have been inconsistent and difficult to reproduce.

The purpose of this paper is to report new paleomagnetic results from the Siwalik Group in the remote far western District of Nepal (Fig. 1), where no previous studies have been attempted. A well-exposed 3400 m-thick section at Khutia Khola was chosen for detailed magnetostratigraphic study in order to establish a template for future studies and to complement contemporaneous studies of the structure, sedimentology, provenance, and geochemistry of the Siwalik Group in western Nepal (Quade et al., 1997; DeCelles et al., 1998a, 1998b).

From previously published work it is clear that there has been no uniformity in sampling strategies among paleomagnetists working in the Siwalik Group in Nepal. Previously sampled rock types have included sandstone (Appel et al., 1991; Gautam and Appel, 1994), laminated siltstone (Harrison et al., 1993), and undifferentiated

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mudrocks. Given the diversity of previously employed sampling strategies and the inconsistency of the paleomagnetic results, we resampled previously worked sections in central Nepal in order to establish which rock types, if any, provide reliable paleomagnetic results for magnetostratigraphy. In addition to the Khutia Khola section (which had never been sampled), we sampled the Siwalik Group at Tinau Khola and Surai Khola, and older (early Miocene) foreland basin deposits in the Dumri Formation along the Tansen road and at Swat Khola (Fig. 1).

SIWALIK GROUP GEOLOGY AT KHUTIA KHOLA

In Nepal, the Siwalik Group crops out between two major thrust faults, the Main Boundary thrust to the north and the Main Frontal thrust along the topographic break between the mountains and the Gangetic plain to the south (Fig. 1). Homoclinal exposures of very thick (>4500 m) sections of the Siwalik Group are present at elevations of 600–2200 m along the many deep canyons that cut the frontal fold-thrust belt (Harrison et al., 1993, Quade et al., 1995, DeCelles et al., 1998a). The study area at Khutia Khola is located 18 km north of Dhangadhi in far western Nepal (Fig. 1). The Siwalik Group is composed entirely of fluvial deposits, and is informally divided into lower, middle, and upper members (Tokuoka et al., 1986; Quade et al., 1995), which are briefly described in the following. The members are recognizable in the Siwaliks all across Nepal, although there is no firm published evidence on whether these members are isochronous or diachronous (see discussion in Quade et al., 1995). Detailed sedimentological descriptions and interpretations of the Siwalik Group were given in Quade et al. (1995) and DeCelles et al. (1998a).

Lower Member

The base of the section at Khutia Khola is covered by young alluvium, and 862 m of lower Siwalik rocks are exposed. Sandstone bodies throughout the Siwalik Group are characterized by two types: lenticular channel deposits, and tabular crevasse-splay deposits (DeCelles et al., 1998a). The lower Siwalik channel sandstone bodies are reddish-brown to brownish-gray and range in thickness from 1 to 10 m. Mudrocks at all levels in the Siwalik Group in Nepal are composed of laminated siltstones and paleosols (Fig. 2), which occur on average every ~13 m (Quade et al., 1995). As in other parts of Nepal, the lower Siwalik paleosols in Khutia Khola are pale yellow to reddish-brown, and less often greenish-gray. The bright colors of these paleosols, coupled with the relatively thin sandstone beds, are characteristic of the lower Siwalik member throughout Nepal. The paleosols contain numerous pedogenic carbonate and iron oxide nodules. The laminated siltstones are mainly gray and brownish-gray, and are characterized by plane-parallel laminations and local ripple cross-laminations (Fig. 2B). Many units of the laminated siltstone contain leaf impressions. The siltstones often cap paleosols and are most common just below the basal erosional surfaces of channel sandstones. Very thin (~2 cm), dark gray, carbonaceous beds are also commonly associated with the laminated siltstones. Willis (1993) noted a similar association of lithofacies in the Siwalik Group of

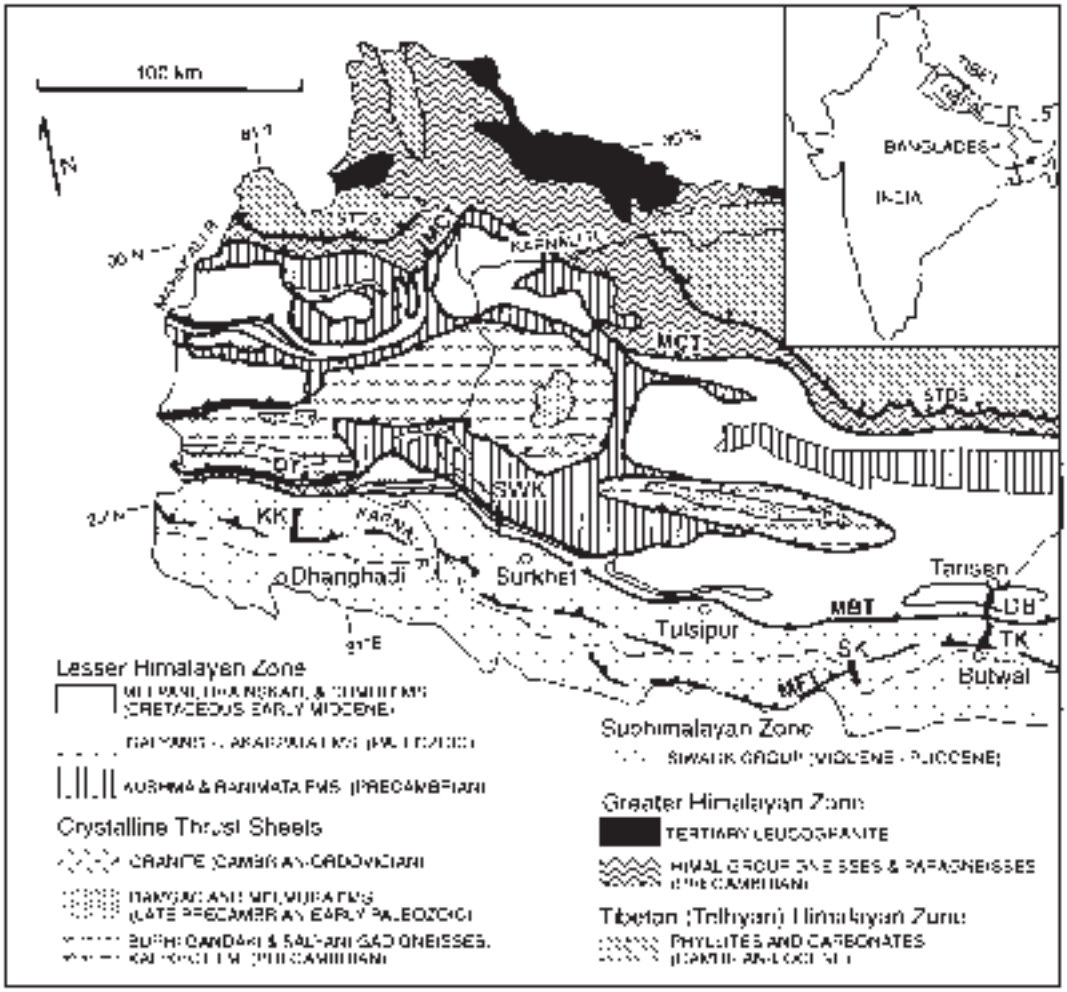


Figure 1. Generalized geological map of far western Nepal modified from DeCelles et al. (1998a). Paleomagnetic sampling locations are Khutia Khola (KK), Swat Khola (SWK), Surai Khola (SK), Tinau Khola (TK), Tinau Khola at Dumri bridge (DB), and major faults are Main Central thrust (MCT), Dadeldhura thrust (DT), Ramgarh thrust (RT), Main Boundary thrust (MBT), Main Frontal thrust (MFT), and South Tibetan detachment system (STDS). Inset shows location of map area in context of the Himalayan orogenic belt.

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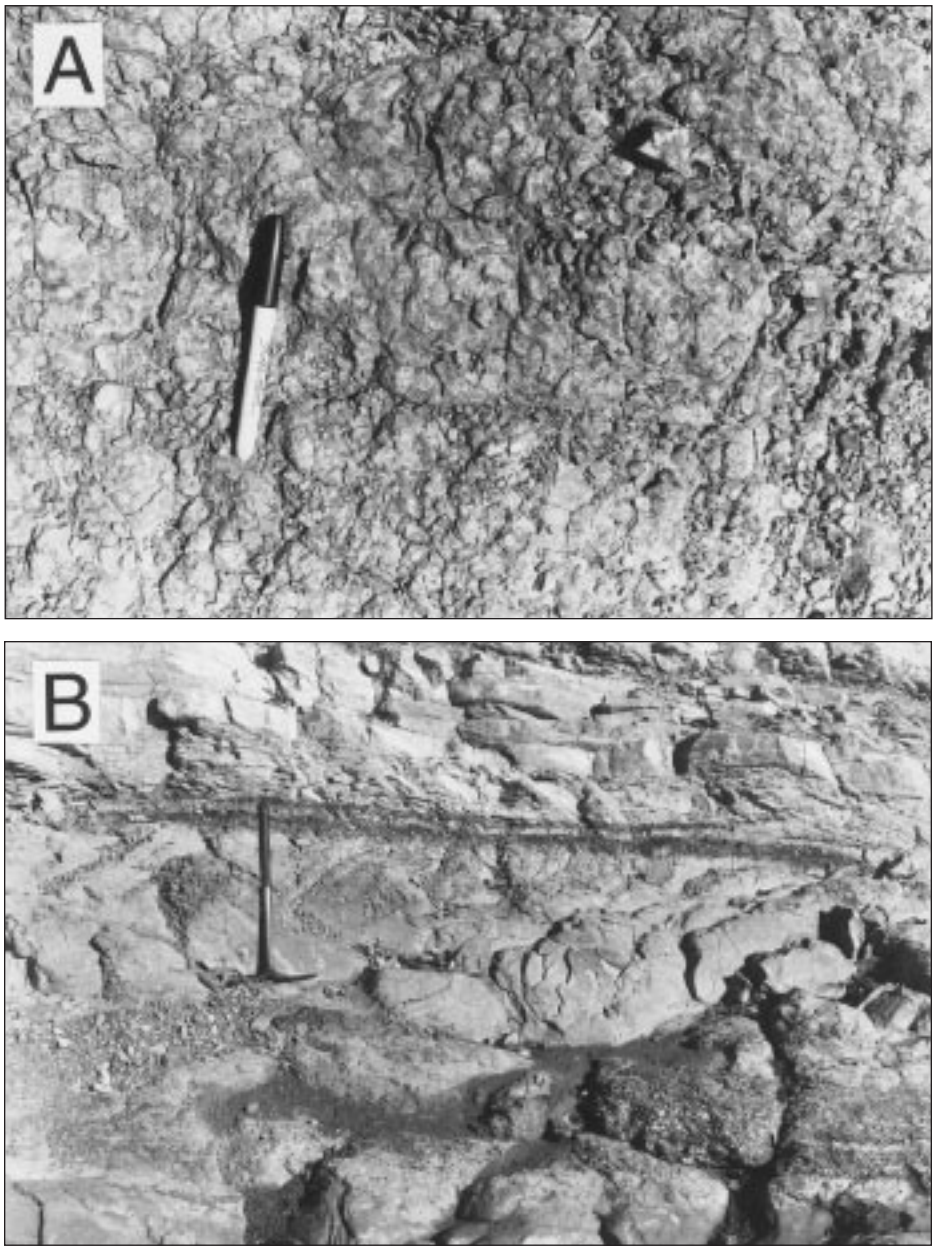


Figure 2. (A) Oxidized, silty paleosol in the lower Siwalik member at Khutia Khola. Note the spheroidal weathering, mottling, and complete absence of primary sedimentary laminations. (B) Example of laminated siltstone lithofacies in the middle Siwalik member at Khutia Khola, resting on top of a massive, bioturbated paleosol. Pick is ~75 cm long; the top of the handle is in two organic-rich zones at the base of the laminated siltstone above the paleosol. The laminated siltstone yielded the best quality paleomagnetic results.

northern Pakistan. The laminated siltstones are interpreted as paludal swamp and backwater deposits that were deposited on low-lying areas of the flood plain (DeCelles et al., 1998a).

Middle and Upper Siwalik Members

The middle Siwalik Group in Khutia Khola is 2468 m thick, and is dominated by medium- to coarse-grained, micaceous, quartzolithic sandstones that have a salt-and-pepper appearance. Individual sandstone bodies range in thickness to ~40 m. The sandstones in the upper part of the section are more friable and contain quartzite pebbles. These large sandstone bodies have been interpreted as major fluvial channel deposits (Tokuoka et al., 1986; DeCelles et al., 1998a). Laminated siltstone in the middle Siwalik member tends to be darker gray than lower in the section, and coal and lignite are common. Middle Siwalik paleosols are entirely different from lower Siwalik paleosols in being typically more gray, green, and yellow, and gleyed paleosols are more common (Quade et al., 1995). Middle Siwalik paleosols also contain abundant humic clay. In the upper Siwalik member, mudrocks (including paleosols) and sandstones are increasingly replaced by thick beds of imbricated, clast-supported conglomerate that are interpreted as gravelly fluvial channel deposits (DeCelles et al., 1998a). The upper member is at least 1000 m thick, but poor exposure inhibits detailed study. At Khutia Khola the top of the section is truncated by a thrust fault, and the upper part of the upper member is largely unexposed.

SAMPLING AND LABORATORY METHODS

At Khutia Khola the Siwalik section is about 3400 m thick, but only the lower 2423 m of the section were sufficiently exposed for continuous paleomagnetic sampling. The base of the sampled section is located at 28°54′N; 80°40′E. The main portion of the sampled section proceeds north along the main drainage for ~8 km; a portion of the sampling was also done in a side canyon (Toli Gad). Of the exposed section, 862 m belong to the lower Siwalik member and 1561 m to the middle member. At least three samples were collected from each of 168 sedimentary beds, which we refer to as paleomagnetic sites. Most of the samples were cored in situ with standard gasoline-driven drilling equipment following protocols described by Butler (1992). About 50 sites were in clay-rich mudrocks that were too poorly indurated for drilling; these were sampled using a car battery–driven electric drill and attached, specially designed rock cutter (Ellwood et al., 1993).

Following sample preparation, all paleomagnetic samples were stored, measured, and thermally demagnetized in a magnetically shielded room at the University of Arizona; the average field intensity was <200 nanotesla (nT). Measurement of natural remanent magnetization (NRM) was done using a three-axis cryogenic magnetometer (2G Model 755R). The initial NRM intensities were found to range between 10^{-1} and 10^{-3} A/m. After initial NRM measurements, samples were subjected to progressive thermal demagnetization. The temperature controllers used in demagnetization are capable of attaining a desired temperature with precision of ±3 °C. Temperature steps (10–16) to 680 °C were employed for progressive thermal demagnetization.

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COMPONENTS OF REMANENT MAGNETISM IN SIWALIK DEPOSITS

Abundant prior research on paleomagnetism of Siwalik sediments established thermal demagnetization as the superior method for isolation of the components of NRM (Tauxe and Opdyke, 1982; Johnson et al., 1985; Appel et al., 1991; Harrison et al., 1993; Gautam and Appel, 1994). In addition to the paleomagnetic study of Siwalik sediments at Khutia Khola fully reported in this paper, we have analyzed the paleomagnetism of the Siwalik Group at Tinau Khola and Surai Khola, and the Dumri Formation along Tansen road (Fig. 1). Complete results will be reported elsewhere, but we report here observations on lithologic control of paleomagnetism. Our pilot set of samples focused on three rock types: sandstone, laminated siltstone, and paleosol (i.e., massive, pedoturbated and bioturbated siltstone).

Our results show that paleosols (Fig. 3, A–C) and sandstones (Fig. 3, D–F) have erratic thermal demagnetization behaviors. The few sandy siltstones that were sampled showed more coherent behavior. In contrast, thermal demagnetization of samples from laminated siltstone (Fig. 4) displayed coherent demagnetization trajectories, many without significant directional changes during thermal demagnetization. Our results show that wherever possible in the Siwalik Group of Nepal, laminated siltstone should be sampled for paleomagnetic studies, and we have exclusively focused the remainder of our paleomagnetic sampling on laminated siltstones. Drawing the distinction in the field between laminated siltstone and paleosols is essential, even though paleosols have only recently been recognized in the Siwalik Group in Nepal (Harrison et al., 1993; Quade et al., 1995). Criteria for field recognition of paleosols include extensive bioturbation producing spheroidal weathering patterns, mottling, root traces, burrows, and Fe/Mn-oxide and carbonate nodules (Fig. 2A). In many places thin beds of siltstone suitable for paleomagnetic sampling cap paleosols and underlie thick sandstone beds.

Tauxe and Badgley (1988) and Tauxe et al. (1990) provided detailed analyses of rock magnetism in Siwalik Group sediments of the Dhok Pathan Formation in northern Pakistan. Although we lack detailed petrographic and anisotropy of magnetic susceptibility information necessary for detailed comparison between the Siwalik sediments of Nepal and those from Pakistan, some useful comparisons and contrasts can be made. In both regions, laminated and unaltered rocks (termed gray sediments in Pakistan) often exhibit coherent demagnetization behaviors with unblocking temperatures extending to 680 °C and often concentrated between 600 °C and 680 °C (Fig. 4). These high unblocking temperature components are almost certainly carried by specular hematite, which acquired a magnetization at or soon after deposition and have been little disturbed since deposition. Tauxe et al. (1990) demonstrated that gray sediments retain primary depositional fabric, indicating that the characteristic magnetization originated as a depositional magnetization. Apparently the laminated siltstones of the Siwalik deposits in Nepal are analogous to the gray sediments of Pakistan.

There is a major contrast between paleomagnetic results from paleosols in Siwalik deposits of Nepal compared to those from Pakistan. We find only incoherent demagnetization behavior from paleosols in Nepal (Fig. 3, A–C), whereas Tauxe and Badgley (1988) indicated that the best magnetostratigraphic data from Pakistan are obtained from paleosols that often have high concentrations of pigmentary hematite. We speculate that this difference is due to little or no precipitation of pigmentary hematite in the paleosols of the Nepal Siwalik deposits. Tauxe and Badgley (1988) and Tauxe et al. (1990) documented the presence of both specular and pigmentary hematite in the overbank flood-plain deposits in Pakistan, the most altered of which show extreme features of pedogenic alteration. Red paleosols of Pakistan with coherent thermal demagnetization behaviors contain both specular hematite and red pigmentary hematite concentrated in cracks. Tauxe and Badgley (1988) argued that, because the chemical remanent magnetization (CRM) component carried by the pigmentary hematite is easily isolated and is abundant in the red sediments of the Pakistan Siwalik deposits, these red paleosol deposits provide reliable magnetostratigraphic data. However, Tauxe and Badgley (1988) also observed paleosols in which the detrital specular hematite grains had been randomized by physical disturbance after deposition. Those Pakistan paleosols yielded incoherent thermal demagnetization behaviors similar to our observations for all Siwalik paleosols in Nepal (Fig. 3, A–C). We speculate that paleosols in the Siwalik deposits of Nepal contain specular hematite grains, which have been physically disturbed after deposition but do not contain sufficient pigmentary hematite to have formed a postdepositional CRM. Consistent with this interpretation is the observation that paleosols in the Siwalik Group of Nepal rarely appear as red in outcrop as do the Siwalik Group paleosols in Pakistan.

PALEOMAGNETISM IN KHUTIA KHOLA DEPOSITS

Following initial NRM measurements, at least three samples from each site at Khutia Khola were subjected to progressive thermal demagnetization. A wide range of thermal demagnetization behaviors was observed, as seen in vector end-point diagrams (Zijderveld, 1967) (Fig. 4). Unblocking temperatures for almost all finely laminated siltstone samples are mainly above 600 °C, indicating that hematite is the dominant carrier of NRM. For most samples, thermal demagnetization reveals essentially univectorial decay of NRM toward the origin of the vector endpoint diagram (e.g., Fig. 4, B–E). A minority of samples showed evidence of a low unblocking temperature component approximately aligned with the present geomagnetic field. Principal component analysis (Kirschvink, 1980) of NRM remaining at four to six successive temperature steps generally above 600 °C was used to determine the characteristic remanent magnetism (ChRM) component. For samples with univectorial NRM, the maximum angular deviation (MAD) was often <5°, indicating a precise determination of the ChRM. For samples from 27 sites and occasional samples from other sites in the collection, thermal demagnetization resulted in erratic behavior from which no ChRM direction could be determined. These samples and samples for which principal component analysis yielded MAD >15° were rejected from further analysis.

Following determination of sample ChRM directions, site-mean ChRM directions were determined using methods of Fisher (1953) and examined using the test for randomness of Watson (1956). Sites with three or more samples yielding ChRM directions that are nonrandom at the 5% significance level and with site-mean 95% confidence limit (α_{95}) <20° are considered the most reliable results and are designated as class A sites. Sites with three or more samples yielding more dispersed ChRM directions but unambiguous polarity are designated class B sites. Sites yielding only two sample ChRM directions with MAD <15° or with two samples yielding well-grouped ChRM directions of unambiguous polarity and a third sample of ambiguous polarity were designated class C sites. Out of 168 sampled sites, 68 sites are class A, 23 sites are class B, 23 are class C, and 27 sites yield ambiguous results and were rejected from further analysis. Site-mean directions are listed in Table DR1 in the GSA Data Repository.[1]

Site-mean ChRM directions for class A sites are plotted on an equal-area projection in Figure 5A. These directions pass the reversal test of McFadden and McElhinny (1990) with a B classification. The null hypothesis that the opposite polarity sets of ChRM site-mean directions were drawn from populations with means

[1]GSA Data Repository item 200031, site-mean directions, is available on the Web at http://www.geosociety.org/pubs/drpint.htm. Requests may also be sent to Documents Secretary, GSA, P.O. Box 9140, Boulder, CO 80301; e-mail: editing@geosociety.org.

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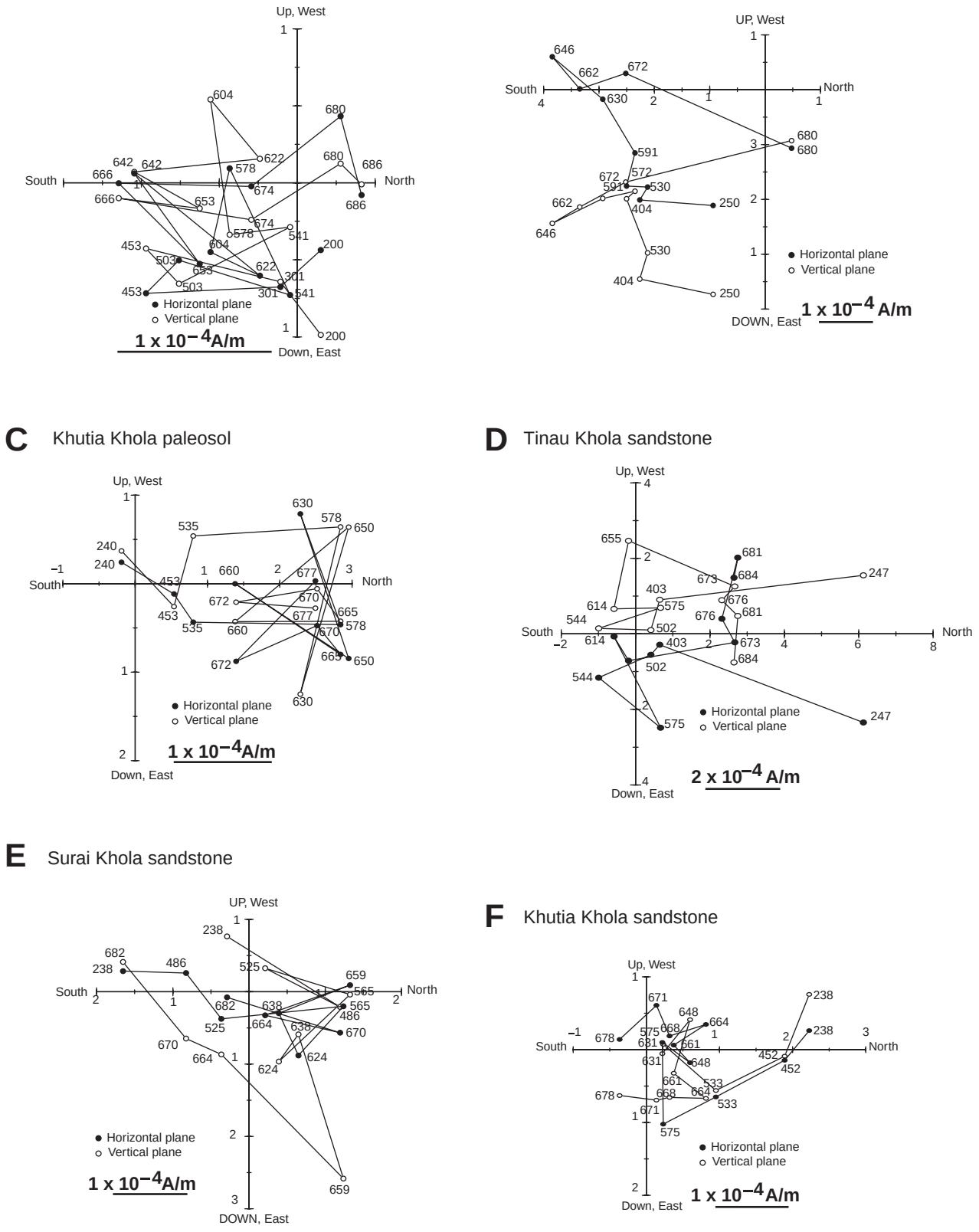


Figure 3. Vector-component diagrams of thermally demagnetized behavior for paleosols and sandstones. Open circles are projections onto vertical plane, and filled circles are projections onto horizontal plane. Numbers adjacent to data points indicate temperature (in °C). (A) Sample of lower Siwalik paleosol from Tinau Khola section. (B) Middle Siwalik paleosol sample from Surai Khola. (C) Sandy paleosol sample from the lower Siwalik member at Khutia Khola. (D) Fine- to medium-grained middle Siwalik sandstone sample from Tinau Khola section. (E) Fine-grained sandstone sample of the lower Siwalik member from Surai Khola. (F) Very fine-grained silty sandstone sample from the lower Siwalik member at Khutia Khola.

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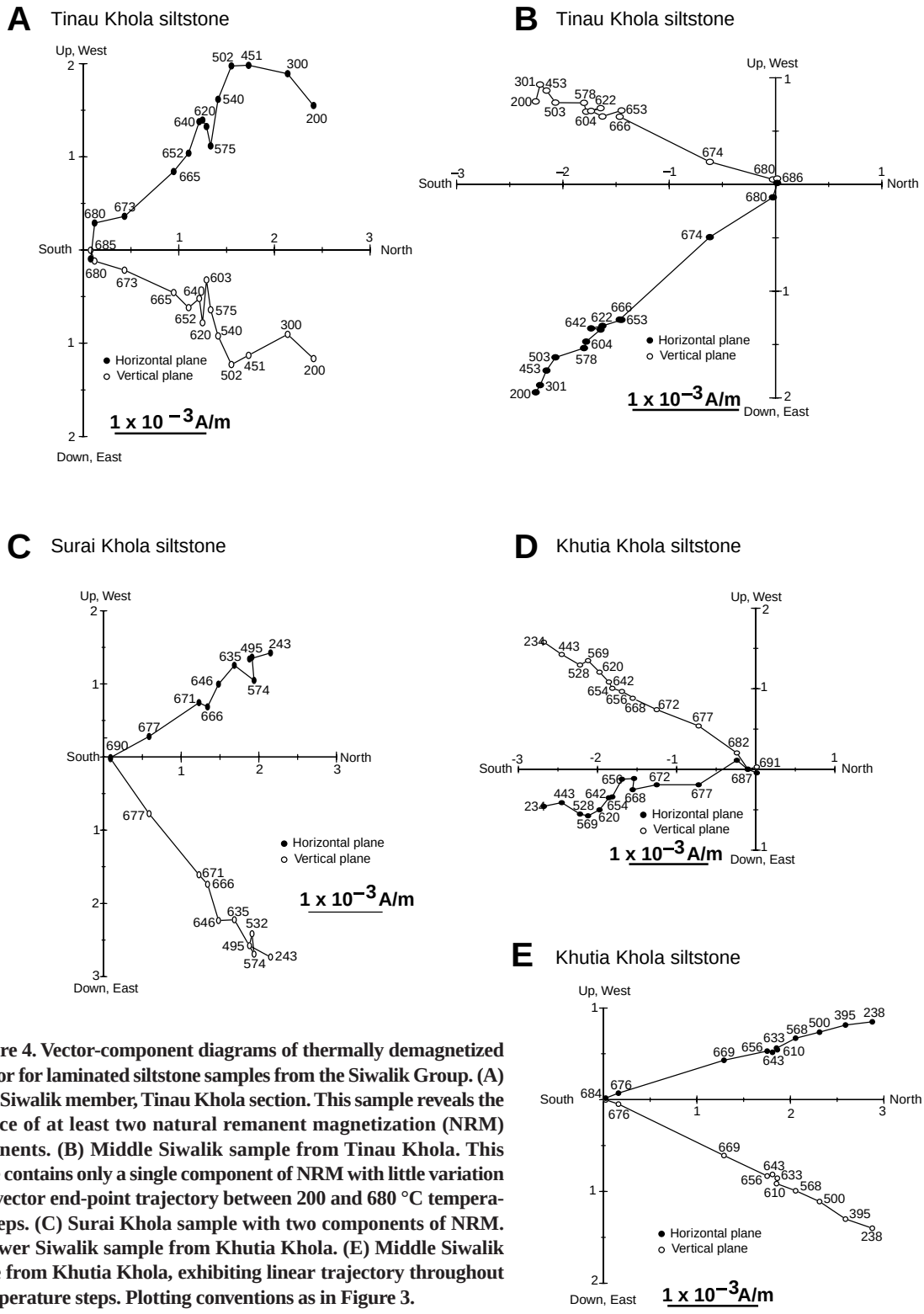


Figure 4. Vector-component diagrams of thermally demagnetized behavior for laminated siltstone samples from the Siwalik Group. (A) Lower Siwalik member, Tinau Khola section. This sample reveals the presence of at least two natural remanent magnetization (NRM) components. (B) Middle Siwalik sample from Tinau Khola. This sample contains only a single component of NRM with little variation in the vector end-point trajectory between 200 and 680 °C temperature steps. (C) Surai Khola sample with two components of NRM. (D) Lower Siwalik sample from Khutia Khola. (E) Middle Siwalik sample from Khutia Khola, exhibiting linear trajectory throughout all temperature steps. Plotting conventions as in Figure 3.

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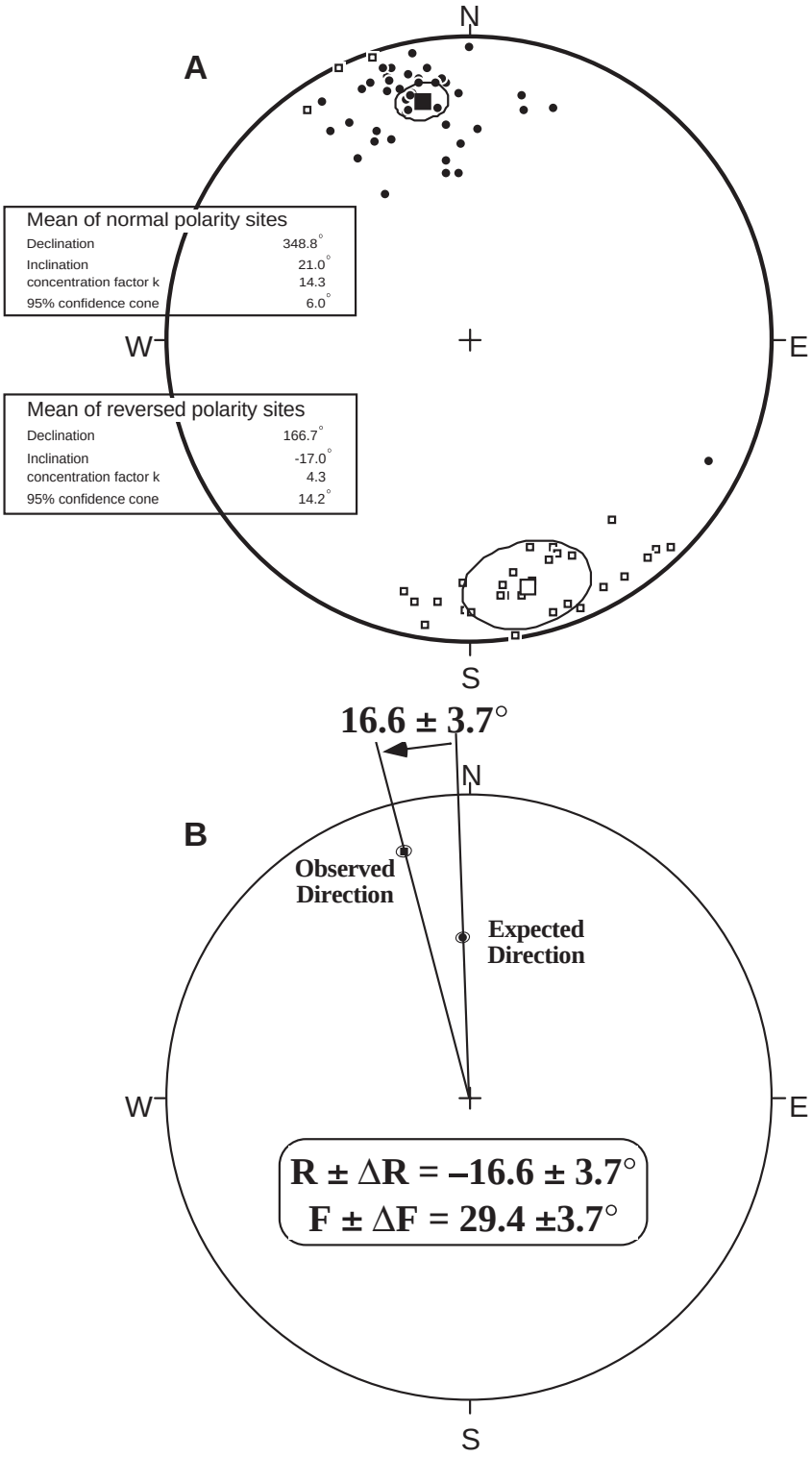


Figure 5. (A) Equal area projection of site-mean characteristic remanent magnetism (ChRM) directions (stratigraphic coordinates) for class A sites from Khutia Khola. Open squares are directions in upper hemisphere; filled circles are directions in lower hemisphere. (B) Equal area projection of overall mean direction calculated from class A ChRM site-mean directions (tilt corrected) in the Khutia Khola section, illustrating the counterclockwise declination discordance of 12.4° and inclination shallowing of 29.4°.

180° apart is accepted at >95% confidence level. Mean directions of the opposing polarity sites have a critical angle of 8.1°, whereas the observed angle between the mean of the normal polarity sites and the antipode of the mean of the reversed polarity sites is 5.6° (Fig. 5A).

The sampled section at Khutia Khola is part of a north-dipping homoclinal sequence. Yet there are sufficient variations in bedding attitude to permit a fold test of the paleomagnetic directions. After deletion of two aberrant ChRM directions, the 66 class A site-mean directions have an in situ k value of 16.57 and a SCOS (measure of correlation between magnetization direction and bedding attitude [McFadden, 1990]) value of 8.1. Upon restoration of beds to horizontal, the k value is 24.10 and SCOS is 2.86, indicating passage of both the McElhinny (1964) and McFadden (1990) versions of the fold test at the 95% confidence level. (Although minimum SCOS value is obtained at 50% unfolding and maximum k is obtained at 75% unfolding, neither statistic is significantly removed from the 100% unfolding value.)

Several lines of evidence suggest that the ChRM is a primary or near primary magnetization acquired ≤1 k.y. following deposition. The generally good precision of determination of sample ChRM directions from thermal demagnetization and the good within-site grouping of those directions suggest a fairly uncomplicated NRM acquisition process. The angular dispersion (angular standard deviation) of virtual geomagnetic poles (VGPs) for the class A sites is 14.4°, which compares closely to the 13.5° value compiled for rocks with ages to 5 Ma between lat 20° and 30° by Merrill and McElhinny (1983). This observation suggests complete sampling of geomagnetic secular variation and indicates that the time interval of NRM acquisition was short compared with the few thousands of years of dominant periodicities of secular variation. Tauxe and Kent (1984) provided strong arguments that the NRM of similar sedimentary rocks in the Siwalik Group of Pakistan was a depositional remanent magnetism carried by hematite and was acquired essentially at the time of deposition. Our observation of significant shallowing of the mean inclination of ChRM compared with the expected geomagnetic field direction at the time of deposition (see following) provides further evidence for a depositional origin of the ChRM in the Siwalik Group of Khutia Khola.

TECTONIC ANALYSIS OF PALEOMAGNETIC DIRECTIONS

The overall mean direction of tilt-corrected site-mean ChRM directions for the 74 class A sites from Khutia Khola is: inclination = 17.0°;

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declination = 345.5°; k = 20.69; α_{95} = 3.7°. Besse and Courtillot (1991) provided a 10 Ma reference paleomagnetic pole for India located at 87.8°N, 203.3°E with confidence limit A_{95} = 2.3°. This reference pole predicts an expected magnetic field direction at the Khutia Khola collecting locality of: inclination = 46.4° ± 2.8°; declination = 2.1° ± 2.6°. Using the methods of Beck (1980) and Demarest (1983), a comparison of the expected and observed directions yields flattening of inclination, F ± ΔF = 29.4° ± 3.7°, and vertical-axis rotation of declination, R ± ΔR = –16.6° ± 3.7° (Fig. 5B).

Tauxe and Kent (1984) showed that modern Soan River sediments of Pakistan acquired an inclination of depositional remanent magnetism that was systematically shallower than the known geomagnetic field direction. This observation was interpreted to be the result of plate-like detrital hematite particles settling with long axes subparallel to bedding. We have strong evidence that the ChRM of the Siwalik Group at Khutia Khola is carried by hematite. Thus we interpret the shallowing of mean inclination to result in depositional shallowing of the acquired magnetization direction, perhaps also augmented by compaction shallowing of magnetic inclination. We do not believe that the shallow mean inclination has tectonic significance.

We believe the vertical-axis rotation of declination is of tectonic significance. This result indicates a counterclockwise rotation of 16.6° ± 3.7° (Fig. 5B). A similar counterclockwise rotation has been documented in the Siwalik Group of the Potwar area in northern Pakistan (Opdyke et al., 1982). In the Tinau Khola section of central Nepal, Gautam and Appel (1994) also noted a mean declination 20° counterclockwise from the expected declination and interpreted this as a local tectonic effect. However, it is possible that the regionally consistent counterclockwise vertical-axis rotation may have broader tectonic significance. Apparently large portions of Siwalik thrust sheets in Nepal have undergone systematic counterclockwise vertical-axis rotation on the order of 15° with respect to the interior of the Indian plate on which they now rest. This sense of rotation is consistent with rotation of rocks in the eastern half of the Himalayan orogenic salient (e.g., Marshak, 1988).

ESTIMATION OF AGE

The age range of the Siwalik Group in Nepal is generally considered to be between ca. 14 Ma and ca. 1 Ma, based on vertebrate fossils (West et al., 1978, 1991; Corvinus, 1994). We found no fossils in our section that would provide more specific biostratigraphic constraints on age. Of 176 sites, 142 were utilized for magnetostratigraphic analysis. We considered two correlations (Figs. 6 and 7) of our magnetostratigraphic column to the geomagnetic polarity timescale (GPTS) of Cande and Kent (1995) (henceforth designated CK95). In all, 11 normal and 11 reverse polarity zones are observed. A thick normal polarity zone between 770 m and 1345 m is correlated with chron C5n of

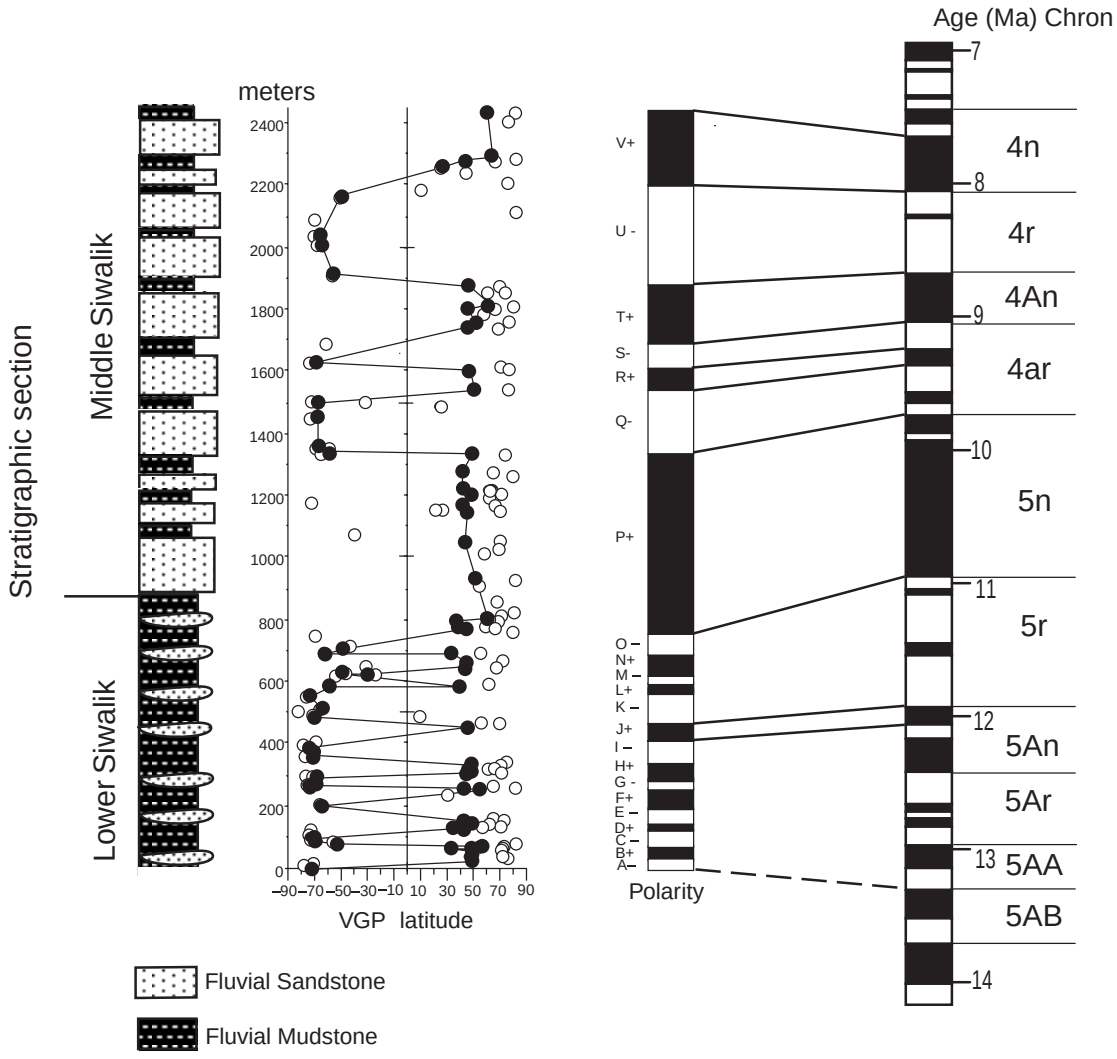


Figure 6. Correlation of the Khutia Khola magnetic polarity stratigraphy to the geomagnetic polarity timescale (GPTS) of Cande and Kent (1995). Open circles are class B data, and solid circles are class A data. VGP is virtual geomagnetic pole.

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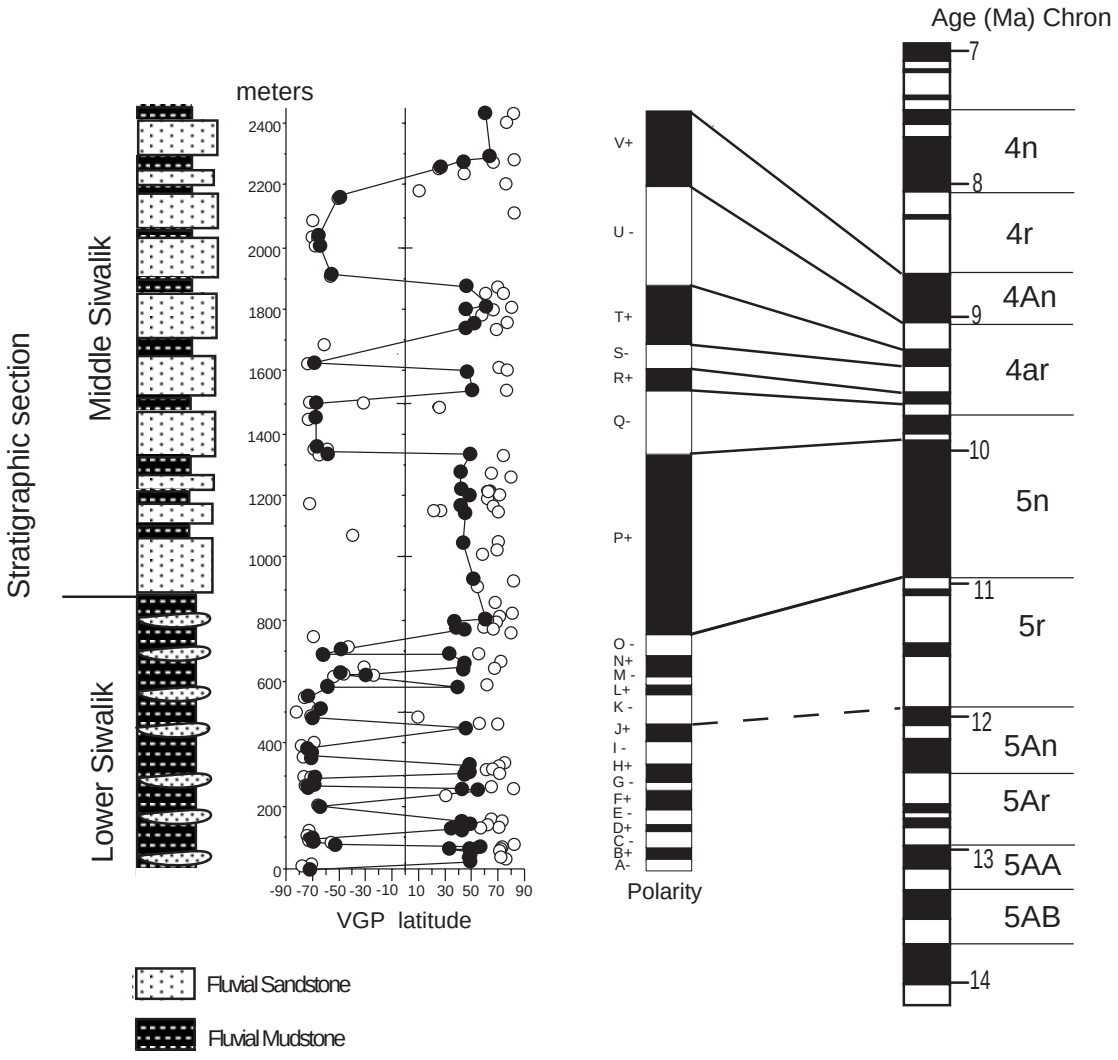


Figure 7. Alternative correlation of the Khutia Khola magnetic polarity stratigraphy to the geomagnetic polarity timescale (GPTS) of Cande and Kent (1995). Symbols and abbreviations as in Figure 6.

CK95. A thick reversed polarity zone with a thin normal zone starting at 1895 m and ending at 2200 m can be correlated with chron C4r.1n in CK95. Except for the lowest 400 m of the section, the match of our paleomagnetic polarity zonation to CK95 appears straightforward (Fig. 6). Using this correlation for the Khutia Khola section, sedimentation began at 13.30 Ma and continued until 7.65 Ma. An alternate, and in our opinion, less convincing correlation to the GPTS of CK95 is shown in Figure 7. According to this correlation, sedimentation began at 11.90 Ma and continued until 8.65 Ma.

Additional age constraints may be provided by carbon isotopic analyses of paleosol carbonates in the section. We analyzed 27 samples of paleosol carbonate following well-established protocols described in Quade et al. (1995). Paleosol carbonate $\delta^{13}C$ values increase elsewhere in Nepal between ca. 7.5 and 7.0 Ma. Our results from Khutia Khola suggest that this shift occurs in the uppermost, undated portion of the measured section (Fig. 8). Markedly enriched $\delta^{13}C$ values typical of paleosol carbonates younger than 7 Ma elsewhere in Nepal occur only in two samples taken from near the top of the measured section. Intermediate $\delta^{13}C$ values (–6.7 to –7.2) were also obtained from samples at ~480 m and ~1750 m. These intermediate values, particularly for the samples from 480 m, are unprecedented for the lower Siwalik member in Nepal, and may signify the presence of C_4 plants earlier in the record than previously documented (see Quade et al., 1995, for evidence from other sections).

ACCUMULATION RATES AND CHRONOLOGY OF HIMALAYAN UNROOFING

The rate of sediment accumulation sharply increases upsection (Fig. 9), using age estimates from our preferred magnetostratigraphic correlation (Fig. 6). Between 13.30 and 10.95 Ma the sediment accumulation rate was ~0.32 mm/yr. Between 10.95 and 9.35 Ma, the rate of sediment accumulation increased to 0.47 mm/yr, and remained this high or higher through the top of the record at 7.65 Ma. As discussed by DeCelles et al. (1998a), the two major structures that were probably active during this time period are the Dadeldhura thrust and a large, hinterland dipping duplex in the Lesser Himalayan zone. One plausible interpretation of the sediment accumulation data (Fig. 9) is that the Dadeldhura thrust sheet was active from 13 to 10.5 Ma, and the Lesser Himalayan duplex became active by ca. 11 Ma. A similar change in sediment accumulation rates throughout much of the Himalayan foreland basin was interpreted by Burbank et al. (1996) to be a result of the emplacement of the Main Boundary thrust sheet. As tempting as it is to interpret small kinks in accumulation curves from foreland basins in terms of individual thrust emplacement events, it should be emphasized that rates of accumulation in well-documented foreland basin systems generally increase smoothly as the fold-thrust belt ap-

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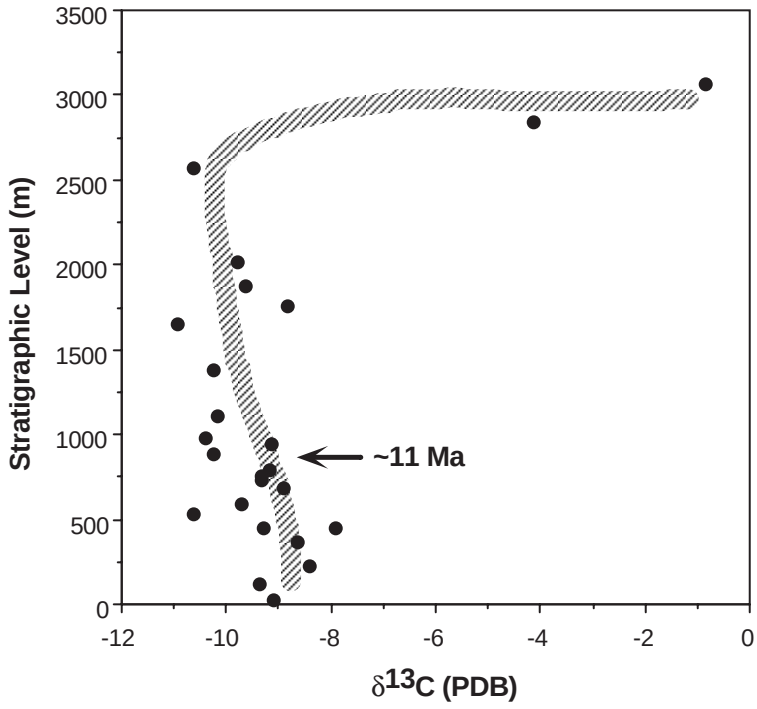


Figure 8. $\delta^{13}C$ values of soil carbonate nodules from Khutia Khola vs. stratigraphic level. PDB—Peedee belemnite.

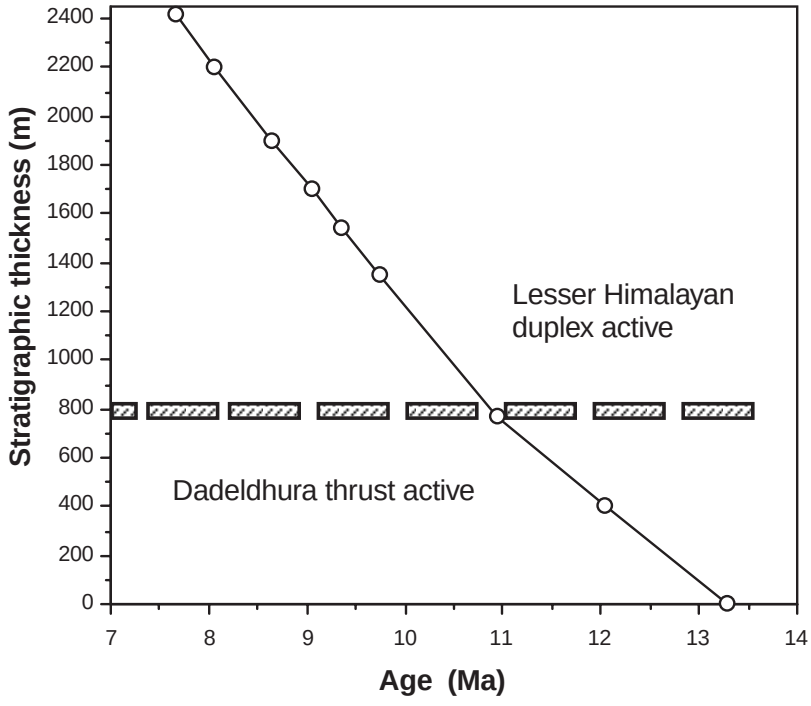


Figure 9. Sediment accumulation rates from the Khutia Khola section, based on the preferred correlation with the geomagnetic polarity timescale (GPTS) of Cande and Kent (1985). Slope of the curve increases markedly ca. 10.9 Ma, based on correlation shown in Figure 6.

proaches a given locality in the foreland basin (Johnson et al., 1986). The long-term (Eocene-Pliocene) history of sediment accumulation from the Nepal portion of the foreland basin shows a progressively increasing rate since early Miocene time (DeCelles et al., 1998b). The increase ca. 11 Ma can be viewed in this context as simply a result of increasing proximity to the orogenic belt as a whole. Multiple thrust displacements, including those within the Lesser Himalayan duplex, helped to bring the topographic load of the rising Himalaya closer to the foreland basin throughout Neogene time. Eventually, the Siwaliks were incorporated into the frontal thrust belt, where they are exposed today.

CONCLUSIONS

Paleomagnetic sampling in the Siwalik Group of Nepal presents a number of challenges. In this paper we present several refinements and improvements over previous studies. Detailed paleomagnetic sampling and analysis demonstrate that paleosols and sandstones, even where very fine grained, do not yield reliable paleomagnetic directions, whereas laminated siltstone yields excellent results. Thermal demagnetization behavior and field tests of paleomagnetic stability suggest that the ChRM is primary or near primary. The correlation of our magnetic polarity zonation with the geomagnetic polarity timescale of Cande and Kent (1995) argues for early acquisition of the ChRM by detrital hematite.

We interpret the shallowing of mean ChRM inclination to be the result of depositional shallowing of the paleomagnetic direction, perhaps also augmented by compaction shallowing. It is unlikely that the shallow mean inclination has tectonic significance. The counterclockwise declination discordance of 16.6° suggests vertical-axis rotation of the Siwalik Group in this region. The apparent regional consistency of this rotation, as indicated by previous studies, suggests vertical-axis rotation of the Siwalik Group in the eastern limb of the main Himalayan orogenic arc. At Khutia Khola, the magnetostratigraphic dating is supported by the stratigraphic level of a major shift in $\delta^{13}C$ values of paleosol carbonates. The shift in $\delta^{13}C$, which has been documented in several other Siwalik sections in Nepal as ca. 7.0–7.5 Ma, takes place above the top of the dated portion of the Khutia Khola section. The dated portion (>7.65 Ma) of the section demonstrates the dominance of C_3 plants. High $\delta^{13}C$ values in the uppermost, undated portion (<7.65 Ma) of the section denote the presence of C_4 plants. The rate of sediment accumulation in the dated part of the section increased sharply after 10.95 Ma. The increase in accumulation rate is best viewed as part of a long-term (>20 m.y.)

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continuum of steadily increasing sediment supply and subsidence rate in the foredeep of the Himalayan foreland basin system, driven mainly by increasing proximity to the southward-migrating fold-thrust belt.

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